How vme-insight-plugin works

The plugin adds a **vme-insight** tab to VME's host, VM, instance and cluster pages, and an optional home-dashboard widget. VME only serves the tab: a frame and a link. The browser then loads the dashboard **directly from vme-insight**, so the VME manager never connects to vme-insight, and the plugin holds no passwords or data. Tabs match the right object because vme-insight's collector stores VME's own ids.

ADMIN'S BROWSER



VME MANAGER

vme-insight-plugin — in VME

- **Tab providers** for servers (hosts and VMs), instances and clusters
- **Home dashboard** "vme-insight", with a summary widget
- **Launcher** (optional): a vme-insight link on every page
- **show()**: tabs appear only when the Grafana URL is set, and, if enabled, for roles with the *vme-insight* permission
- **Returns**: tab HTML (iframe + link) and a page policy, `frame-src <Grafana origin>`, so VME lets the frame load

WHICH PAGE SHOWS WHICH DASHBOARD

VME page	Tab / item	Opens (VME id passed)
Host	Server tab	`/d/vme-hosts?var-host=<id>`
Virtual machine	Server tab	`/d/vme-vms?var-vm=<id>`
Instance	Instance tab	`/d/vme-vms?var-vm=<each VM>`
Cluster	Cluster tab	`/d/vme-overview?var-cluster=<id>`
Home dashboard	Summary widget	`/d-solo/vme-widget?panelId=…`
Every page (opt.)	Launcher link	Grafana in a new tab, no frame

Tabs add `&kiosk`; "Open in vme-insight ↗" opens the same view full size. Host or VM is decided from VME's server type, or the plugin's *Treat as hosts / VMs* settings.

collector reads the VME API (read-only account), separately from the plugin

VME-INSIGHT HOST (DOCKER)

https front end — Caddy · :3443

- **TLS**: the browser must already trust the certificate. A frame can't show a warning, so an untrusted one leaves the tab blank.
- **Allowlist** (`HTTPS_ALLOW`): only admin networks; others get 403
- Reverse proxy to Grafana

Grafana — 11.5

- **Anonymous Viewer** (`GRAFANA_ANONYMOUS`): read-only, no sign-in inside the frame
- **Embedding allowed** (`GRAFANA_ALLOW_EMBEDDING`)
- `GRAFANA_ROOT_URL` = the plugin's Grafana URL, so links stay on one address
- Dashboards filtered by the VME id in the URL

TimescaleDB · Prometheus

History, inventory and lifecycle (SQL); exporter metrics (PromQL). Stored with VME's own host, VM and cluster ids.

collector

Polls the VME API every minute with a read-only account. Separate from the plugin; doesn't involve the browser.

1. An admin opens a host, VM, instance or cluster page in VME.
2. The plugin returns the tab (an iframe and a link) and a `frame-src` policy that lets VME's page show it.
3. The browser loads the dashboard itself, from vme-insight over https, e.g. `/d/vme-hosts?var-host=42`.
4. The front end checks the certificate is trusted and the network is allowed, then passes the request to Grafana.
5. Grafana, as a read-only anonymous Viewer, queries the stored data for that VME id and draws the dashboard.

PLUGIN SETTINGS

Administration → Integrations → Plugins → vme-insight (pencil icon)

vme-insight Grafana URL	The https address users browse to, e.g. `https://vme-insight.example.com:3443`. Unset = no tabs.
Grafana organization, theme	Optional: organization id; light or dark
Tab height	Pixels, default 1400
Home dashboard panel IDs	Tiles from `vme-widget`, default 1,2,3,4,7,8,9,12
Treat as hosts / VMs	Server type codes, for unusual estates
Require permission	Off by default; on = only roles with the vme-insight permission
Show launcher	Off by default; a link on every page

NEEDED IN VME-INSIGHT FIRST

https front end	`docker compose --profile https up -d https`: VME pages are https, and browsers block http frames inside them
Certificate	From a CA your browsers already trust, or a `make-cert.sh host` certificate imported on each admin machine after checking its fingerprint
Who can reach it	`HTTPS_ALLOW` = the networks your VME admins use
No sign-in	`GRAFANA_ANONYMOUS=true` (read-only Viewer), `GRAFANA_ALLOW_EMBEDDING=true`
One address	`GRAFANA_ROOT_URL` = the plugin's Grafana URL (by IP? also `HTTPS_DEFAULT_SNI`)

Step by step: docs/VME-PLUGIN.md. vme-insight works without the plugin; the plugin needs vme-insight.

IF THE TAB DOESN'T SHOW THE DASHBOARD

No vme-insight tab	Grafana URL not set, or *Require permission* on and the role lacks it
Blank tab, "might be having issues"	This browser doesn't trust the certificate (on Windows: imported by double-click, as an intermediate)
"This content is blocked"	Page policy: check the URL setting is saved, reload; plugin 0.1.1 or later
Forbidden (403)	This network isn't in `HTTPS_ALLOW`
Grafana sign-in page	Anonymous viewing off, or Grafana not restarted
`ERR_SSL_PROTOCOL_ERROR`	Browsing by IP: set `HTTPS_DEFAULT_SNI`, or use the name
Links jump to another address	Plugin URL and `GRAFANA_ROOT_URL` differ

Community project, not affiliated with HPE · github.com/grahamfairbank/vme-insight · github.com/grahamfairbank/vme-insight-plugin